```
/*Name - Himanshu
 Branch - E&TC (A1)
 Roll No. - 1124
 Operations on Complex numbers*/

#include <iostream>
using namespace std;
class Complex
{
    float real, imag;
public:
    void input()
    {
        cout << "Enter real part: ";
        cin >> real;
        cout << "Enter imaginary part: ";
        cin >> imag;
    }
    void display()
    {
        if (imag >= 0)
            cout << real << " + " << imag << "i" << endl;
        else
            cout << real << " - " << -imag << "i" << endl;
    }
    // Addition
    Complex add(Complex c)
    {
        Complex temp;
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }
    // Subtraction
    Complex subtract(Complex c)
    {
        Complex temp;
        temp.real = real - c.real;
        temp.imag = imag - c.imag;
        return temp;
    }
    // Multiplication
    Complex multiply(Complex c)
    {
        Complex temp;
        temp.real = (real * c.real) - (imag * c.imag);
        temp.imag = (real * c.imag) + (imag * c.real);
        return temp;
    }
};
int main()
{
    Complex c1, c2, result;
    cout << "Enter first complex number:\n";
    c1.input();
    cout << "Enter second complex number:\n";
    c2.input();
    cout << "\nFirst Complex Number: ";
    c1.display();
```

```
    cout << "Second Complex Number: ";
    c2.display();
    result = c1.add(c2);
    cout << "\nAddition: ";
    result.display();
    result = c1.subtract(c2);
    cout << "Subtraction: ";
    result.display();
    result = c1.multiply(c2);
    cout << "Multiplication: ";
    result.display();
    return 0;
}
```

input

```
Enter first complex number:
Enter real part: 8
Enter imaginary part: -4
Enter second complex number:
Enter real part: 5
Enter imaginary part: -2

First Complex Number: 8 - 4i
Second Complex Number: 5 - 2i

Addition: 13 - 6i
Subtraction: 3 - 2i
Multiplication: 32 - 36i


...Program finished with exit code 0
Press ENTER to exit console.
```

```
/*Name - Himanshu
  Branch - E&TC (A1)
  Roll No. - 1124
  program to compute area / volume*/

#include <iostream>
using namespace std;

class Shape
{
public:
    // Area of Circle
    float area(float r)
    {
        return 3.14 * r * r;
    }
    // Area of Rectangle
    float area(float l, float b)
    {
        return l * b;
    }
    // Volume of Cube
    float volume(int side)
    {
        return side * side * side;
    }
    // Volume of Cylinder
    float volume(float r, float h)
    {
        return 3.14 * r * r * h;
    }
};
int main()
{
    Shape s;
    int choice;
    do
    {
        cout << "\n--- MENU ---\n";
        cout << "1. Area of Circle\n";
        cout << "2. Area of Rectangle\n";
        cout << "3. Volume of Cube\n";
        cout << "4. Volume of Cylinder\n";
        cout << "5. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch(choice)
        {
            case 1:
            {
                float r;
                cout << "Enter radius: ";
                cin >> r;
                if(r > 0)
                    cout << "Area of Circle = " << s.area(r);
                else
                    cout << "Invalid input!";
                break;
            }
```

```
        case 2:
        {
            float l, b;
            cout << "Enter length and breadth: ";
            cin >> l >> b;
            if(l > 0 && b > 0)
                cout << "Area of Rectangle = " << s.area(l, b);
            else
                cout << "Invalid input!";
            break;
        }
        case 3:
        {
            int side;
            cout << "Enter side: ";
            cin >> side;
            if(side > 0)
                cout << "Volume of Cube = " << s.volume(side);
            else
                cout << "Invalid input!";
            break;
        }
        case 4:
        {
            float r, h;
            cout << "Enter radius and height: ";
            cin >> r >> h;
            if(r > 0 && h > 0)
                cout << "Volume of Cylinder = " << s.volume(r, h);
            else
                cout << "Invalid input!";
            break;
        }
        case 5:
            cout << "Exiting program...";
            break;
        default:
            cout << "Invalid choice!";
    }
  } while(choice != 5);
  return 0;
}
```

Output

Clear

```
--- MENU ---
1. Area of Circle
2. Area of Rectangle
3. Volume of Cube
4. Volume of Cylinder
5. Exit
Enter your choice: 2
Enter length and breadth: 5
3
Area of Rectangle = 15
```

```
/*Name - Himanshu
  Branch - E&TC (A1)
  Roll No. - 1124
  Vector operations*/

#include <iostream>
using namespace std;
class Vector
{
    int x, y;
public:
    void input()
    {
        cout << "Enter x and y components: ";
        cin >> x >> y;
    }
    void display()
    {
        cout << "(" << x << ", " << y << ")" << endl;
    }
    // Addition
    Vector add(Vector v)
    {
        Vector temp;
        temp.x = x + v.x;
        temp.y = y + v.y;
        return temp;
    }
    // Subtraction
    Vector subtract(Vector v)
    {
        Vector temp;
        temp.x = x - v.x;
        temp.y = y - v.y;
        return temp;
    }
    // Dot Product
    int dotProduct(Vector v)
    {
        return (x * v.x) + (y * v.y);
    }
};
int main()
{
    Vector v1, v2, result;
    cout << "Enter first vector:\n";
    v1.input();
    cout << "Enter second vector:\n";
    v2.input();
    cout << "\nFirst Vector: ";
    v1.display();
    cout << "Second Vector: ";
    v2.display();
    result = v1.add(v2);
    cout << "\nAddition: ";
    result.display();
    result = v1.subtract(v2);
    cout << "Subtraction: ";
    result.display();
```

```
    cout << "Dot Product: " << v1.dotProduct(v2);
    return 0;
}
```

Output

Clear

```
Enter first vector:
Enter x and y components: 3
5
Enter second vector:
Enter x and y components: 5
8

First Vector: (3, 5)
Second Vector: (5, 8)

Addition: (8, 13)
Subtraction: (-2, -3)
Dot Product: 55

=== Code Execution Successful ===
```